



Provide Insights & Performance Analysis in Transport Domain

Graduation Project at CrackNontech

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Agenda



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Problem Statement



Goodcabs, a cab service company, operates in 10 tier-2 cities in India and supports local drivers while providing excellent service to passengers. The company has set ambitious growth and passenger satisfaction targets for 2024.

The Chief of Operations needs immediate insights into key performance metrics, including trip volume, passenger satisfaction, repeat passenger rate, trip distribution, and the balance between new and repeat passengers.

However, due to the analytics manager's unavailability, the task has been assigned to me, to analyze these metrics and deliver actionable insights.

Goal



The goal of this project is to deliver actionable insights on Goodcabs' performance in 10 tier-2 cities by analyzing key metrics,

- **Identify growth opportunities** by understanding trip trends.
- **Boost customer retention** by increasing repeat passenger rates.
- **Optimize operations** by balancing new and repeat passenger engagement.
- **Support strategic decision-making** to achieve 2024 performance targets.

Data Overview



Database	Table Name	Short Description
trips_db	dim_city	Contains city-specific details for location-based analysis.
trips_db	dim_date	Provides date-specific details for time-based analysis (month, weekday/weekend).
trips_db	fact_passenger_summary	Aggregated monthly summary of total, new, and repeat passengers by city.
trips_db	dim_repeat_trip_distribution	Aggregated breakdown of repeat trip behavior by month, city, and trip frequency (up to 10 trips/month).
trips_db	fact_trips	Detailed trip-level data, including distance, fare, and passenger/driver ratings.
targets_db	city_target_passenger_rating	Monthly target average passenger ratings for each city.
targets_db	monthly_target_new_passengers	Monthly targets for new passenger acquisition by city.
targets_db	monthly_target_trips	Monthly targets for total trip volume by city.

- Dataset are available for 6 months duration from January 2024 to June 2024.
- I have created new table named City Category which have info about (tourism and business focused cities)



Dashboard Preview

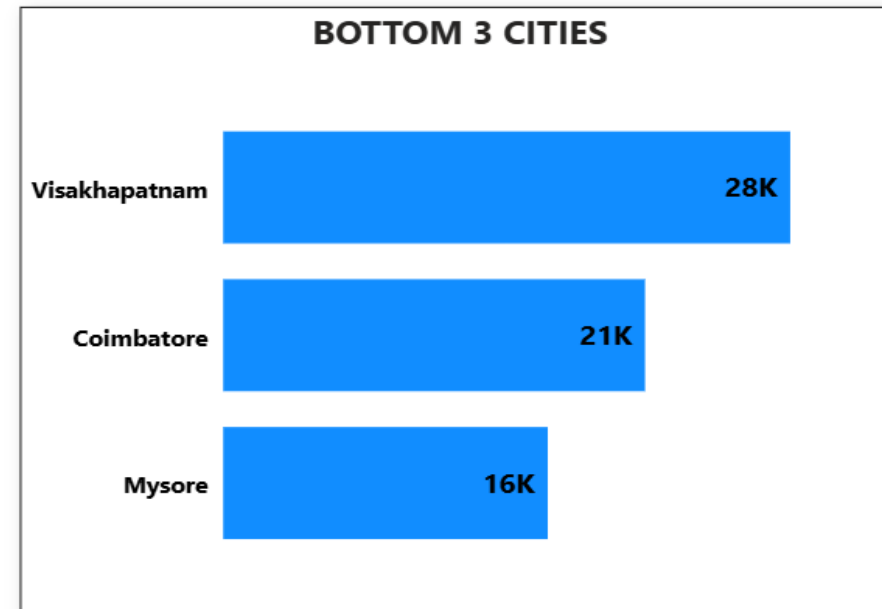
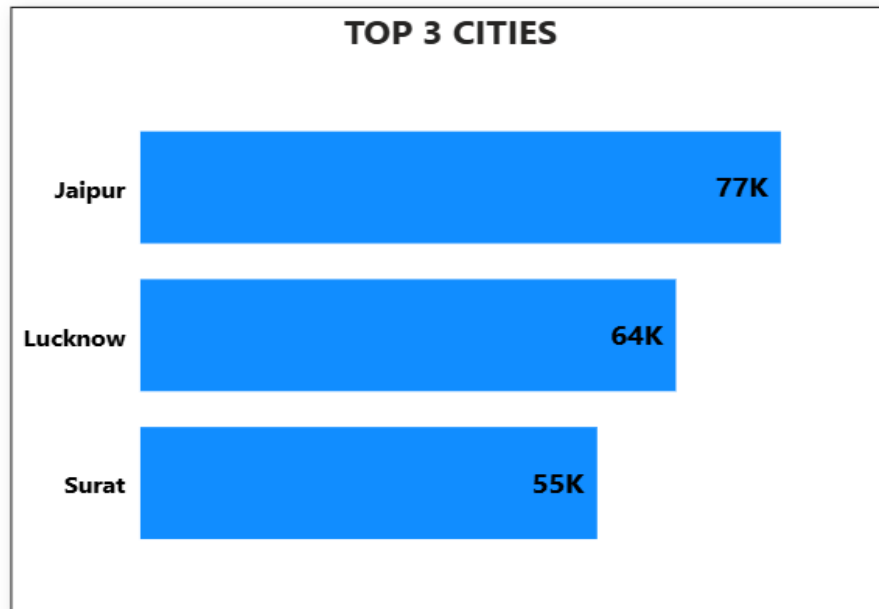
[Live Dashboard Link](#)



Primary Analysis and Insights



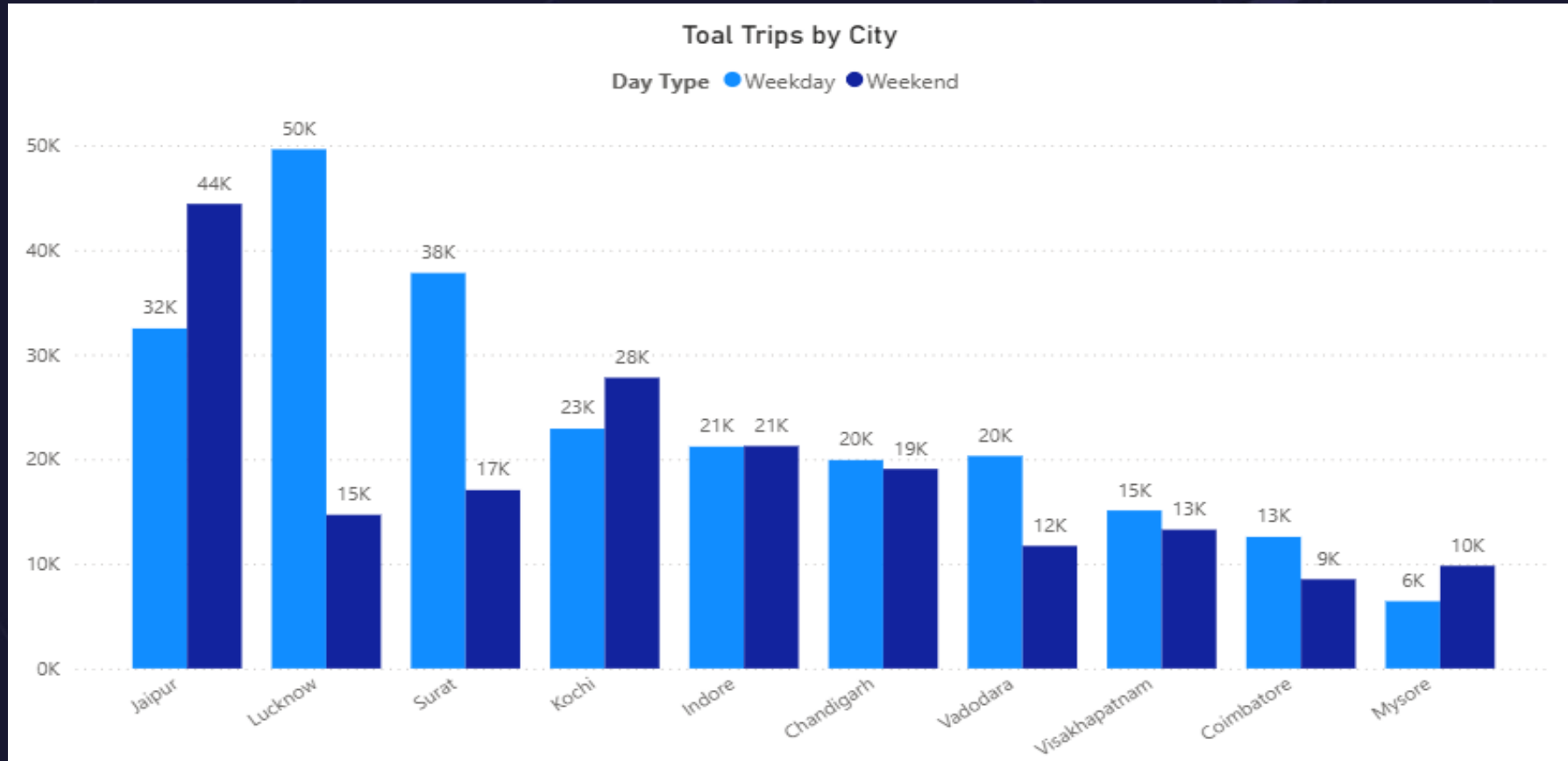
1. Identify the top 3 and bottom 3 cities by total trips over the entire analysis period.



2. Calculate the average fare per trip for each city and compare it with the city's average trip distance. Identify the cities with the highest and lowest average fare per trip to assess pricing efficiency across locations.

city_name	AVG fare_amount	AVG distance_travelled_km	AVG Fare_per_KM
Jaipur	₹ 483.92	30.02	₹ 16.25
Mysore	₹ 249.71	16.50	₹ 15.40
Kochi	₹ 335.25	24.07	₹ 14.13
Visakhapatnam	₹ 282.67	22.55	₹ 12.70
Chandigarh	₹ 283.69	23.52	₹ 12.18
Lucknow	₹ 147.18	12.51	₹ 12.14
Coimbatore	₹ 166.98	14.98	₹ 11.30
Indore	₹ 179.84	16.50	₹ 11.07
Surat	₹ 117.27	11.00	₹ 10.92
Vadodara	₹ 118.57	11.52	₹ 10.54
Total	₹ 254.02	19.13	₹ 12.86

3. Compare the total trips taken on weekdays versus weekends for each city over the six-month period. Identify cities with a strong preference for either weekend or weekday trips to understand demand variations.



4. For each city, evaluate monthly performance against targets for total trips, new passengers, and average passenger ratings. Determine if each metric met, exceeded, or missed the target, and calculate the percentage difference.

Monthly (TA) Status for Total Trips, New Passengers and AVG.Passenger Rating							
City	Category	Trip TA	Trip TD%	New Pass. TA	New Pass. TD%	Avg.Pass.. Rating TA	Avg.Pass. Rating TD%
Chandigarh	Tourism-focused	Missed	-0.05%	Missed	-9.96%	Missed	-0.29%
Coimbatore	Business-focused	Exceeded	0.50%	Exceeded	13.52%	Missed	-4.45%
Indore	Business-focused	Missed	-2.40%	Exceeded	5.41%	Missed	-2.15%
Jaipur	Tourism-focused	Exceeded	13.91%	Missed	-15.08%	Exceeded	4.05%
Kochi	Tourism-focused	Exceeded	2.43%	Missed	-2.16%	Exceeded	0.19%
Lucknow	Business-focused	Missed	-10.70%	Exceeded	4.23%	Missed	-10.49%
Mysore	Tourism-focused	Exceeded	20.28%	Missed	-2.66%	Exceeded	2.37%
Surat	Business-focused	Missed	-3.78%	Exceeded	10.72%	Missed	-8.33%
Vadodara	Business-focused	Missed	-14.60%	Exceeded	2.29%	Missed	-11.85%
Visakhapatnam	Business-focused	Missed	-0.47%	Missed	-5.58%	Missed	-0.79%



Secondary Analysis & Recommendations



1. Factors Influencing Repeat Passenger Rates

➤ Use the data points like repeat passenger count, passenger ratings, city demographics, and pricing data to identify patterns.

➤ **Insights:**

- Higher repeat passenger rates may correlate with better passenger ratings, indicating quality service.
- Competitive pricing might attract more repeat passengers in cities with a price-sensitive demographic.
- Socioeconomic factors: Cities with higher-income groups may prefer premium services, whereas lower-income groups may prefer cost-effective services.
- Lifestyle patterns: Cities with high professional populations (e.g., IT hubs) may show consistent repeat usage during weekdays.

➤ **Recommendation:**

- Focus on improving service quality and offering personalized pricing plans for cities with lower repeat rates.



2. Tourism vs. Business Demand Impact

- Cross-referenced trip data with event calendars (festivals, conferences, tourism seasons) over 6 months.
- **Insights:**
 - **Cities with high tourism activities (e.g., Jaipur, Kochi) show demand spikes during festive seasons or vacations.**
 - **Business-oriented cities (e.g., Lucknow, Surat) show consistent weekday demand.**
- **Recommendation:**
 - Tailor marketing efforts to highlight tourism-related campaigns in high-demand periods.
 - offer flexible services for business travelers e.g., loyalty program, ride packages for conferences.



3. Emerging Mobility Trends and Goodcabs' Adaptation

➤ Evaluate trends like adoption of electric vehicles, passenger preferences for green energy, and emerging mobility needs.

➤ **Insights:**

- **Electric Vehicle (EV) Adoption: Growing preference for green transport in tier-2 cities. EV could reduce operational costs and appeal to environmentally conscious passengers.**
- **Sustainability Focus: Eco-conscious passengers favor ride-sharing and low-emission options.**

➤ **Recommendation:**

- Gradually integrate electric or hybrid vehicles in tier-2 cities to stay competitive and consider offering discounts on eco-friendly Rides.



Ad-Hoc Business Requests

1. Generate a report that displays the total trips, average fare per km, average fare per trip, and the percentage contribution of each city's trips to the overall trips.

	city_name character varying (50) 🔒	total_trips bigint 🔒	avg_fare_per_km numeric 🔒	avg_fare_per_trip numeric 🔒	contribution_to_total_trips numeric 🔒
1	Chandigarh	38981	12.06	283.69	9.15
2	Coimbatore	21104	11.15	166.98	4.96
3	Indore	42456	10.90	179.84	9.97
4	Jaipur	76888	16.12	483.92	18.05
5	Kochi	50702	13.93	335.25	11.90
6	Lucknow	64299	11.76	147.18	15.10
7	Mysore	16238	15.14	249.71	3.81
8	Surat	54843	10.66	117.27	12.88
9	Vadodara	32026	10.29	118.57	7.52
10	Visakhapatnam	28366	12.53	282.67	6.66

2. Generate a report that shows the percentage distribution of repeat passengers by the number of trips they have taken in each city. Calculate the percentage of repeat passengers who took 2 trips, 3 trips, and so on, up to 10 trips. Each column should represent a trip count category, displaying the percentage of repeat passengers who fall into that category out of the total repeat passengers for that city.



	city_name character varying (50) 🔒	2_Trips numeric 🔒	3_Trips numeric 🔒	4_Trips numeric 🔒	5_Trips numeric 🔒	6_Trips numeric 🔒	7_Trips numeric 🔒	8_Trips numeric 🔒	9_Trips numeric 🔒	10_Trips numeric 🔒
1	Kochi	47.67	24.35	11.81	6.48	3.91	2.11	1.65	1.21	0.81
2	Mysore	48.75	24.44	12.73	5.82	4.06	1.76	1.42	0.54	0.47
3	Visakhapatnam	51.25	24.96	9.98	5.44	3.19	1.98	1.39	0.88	0.92
4	Indore	34.34	22.69	13.40	10.34	6.85	5.24	3.26	2.38	1.51
5	Vadodara	9.87	14.17	16.52	18.06	19.08	12.86	5.78	2.05	1.61
6	Jaipur	50.14	20.73	12.12	6.29	4.13	2.52	1.90	1.20	0.97
7	Surat	9.76	14.26	16.55	19.75	18.45	11.89	6.24	1.74	1.35
8	Lucknow	9.66	14.77	16.20	18.42	20.18	11.33	6.43	1.91	1.10
9	Chandigarh	32.31	19.25	15.74	12.21	7.42	5.48	3.47	2.33	1.79
10	Coimbatore	11.21	14.82	15.56	20.62	17.64	10.47	6.15	2.31	1.22

3. Generate a report that calculates the total new passengers for each city and ranks them based on this value. Identify the top 3 cities with the highest number of new passengers as well as the bottom 3 cities with the lowest number of new passengers, categorizing them as "Top 3" or "Bottom 3" accordingly.

	city_name character varying (50) 🔒	total_new_passengers bigint 🔒	city_category text 🔒
1	Jaipur	9682	Top 3
2	Lucknow	9597	Top 3
3	Surat	8638	Top 3
4	Vadodara	4346	Bottom 3
5	Coimbatore	2551	Bottom 3
6	Mysore	1477	Bottom 3



Thank You